Board B — 6-channel coherent digitizer (3 x AD9361, FMC LPC)

AD9361 #1 (CH0, CH1)

File: ad9361_1.kicad_sch

AD9361 #2 (CH2, CH3)

File: ad9361_2.kicad_sch

AD9361 #3 (CH4, CH5)

File: ad9361_3.kicad_sch

RF inputs: SMA, attenuator positions, baluns

File: rf_inputs.kicad_sch

40 MHz reference distribution

File: clock.kicad_sch

Power: 12 V -> 1.8 V -> 3 x 1.3 V low-noise LDOs

File: power.kicad_sch

FMC LPC mezzanine connector

File: fmc.kicad_sch

PWR_FLAG

+12V_FMC

PWR_FLAG

+3V3

PWR_FLAG

GND

Sheet: /
File: board_b.kicad_sch

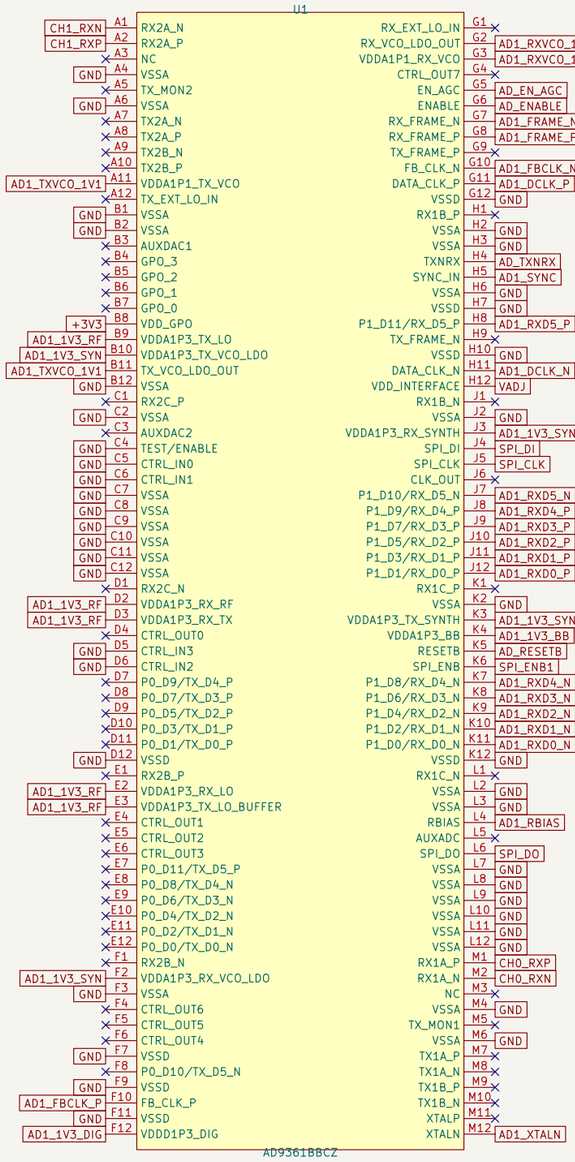
Title: Board B — 6-channel coherent digitizer (3 x AD9361, FMC LPC)

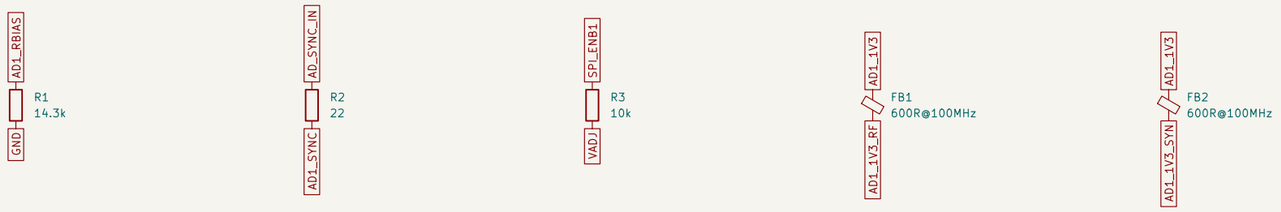
Size: A3 | Date: | Rev:

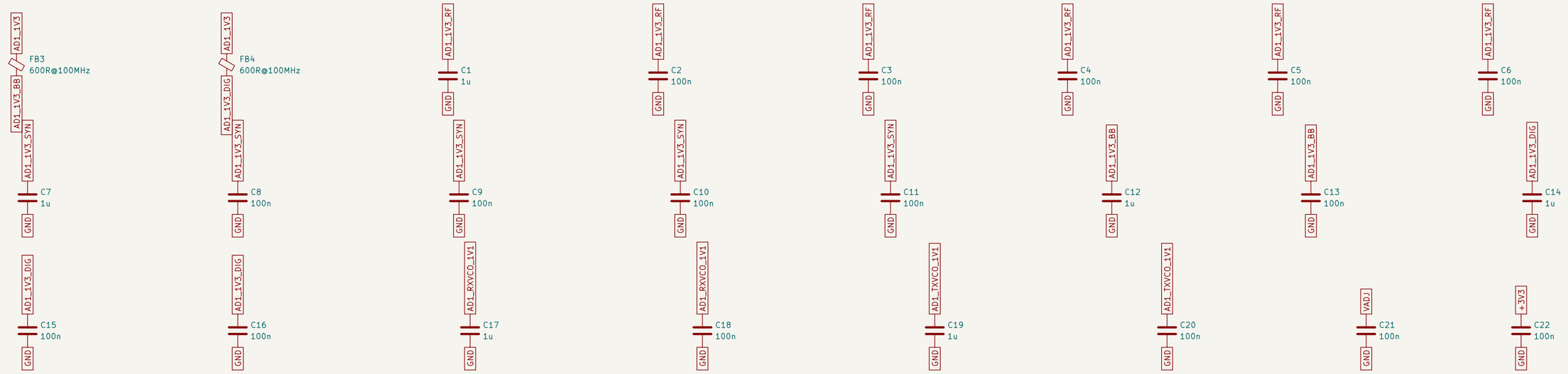
KiCad E.D.A. 10.0.3 | Id: 1/8

AD9361 #1 (CH0, CH1)

Pin-strap/unused-pin handling follows ADI UG-570 and the AD-FMCOMMS2 reference design; verify.







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Passive radar receiver

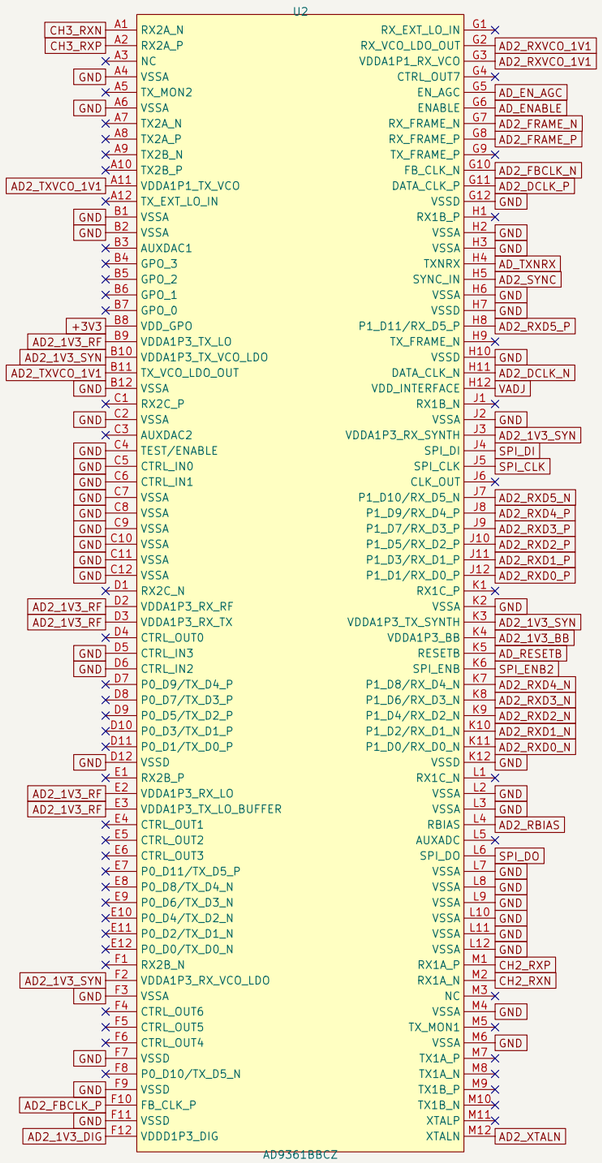
Sheet: /AD9361 #1 (CH0, CH1)/
File: ad9361_1.kicad_sch

Title: AD9361 #1 (CH0, CH1)

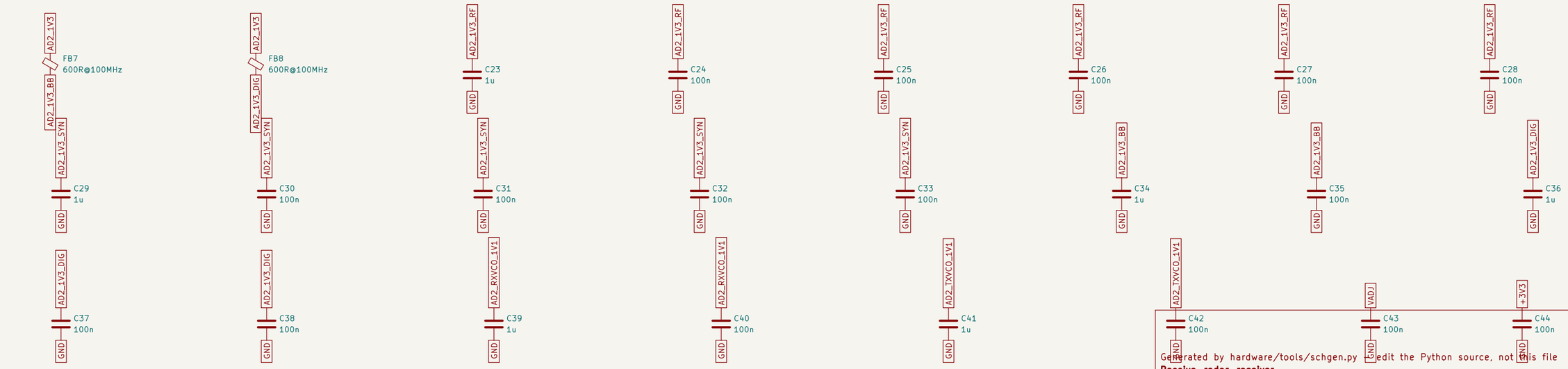
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 2/8

AD9361 #2 (CH2, CH3)

Pin-strap/unused-pin handling follows ADI UG-570 and the AD-FMCOMMS2 reference design; verify.







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Passive radar receiver

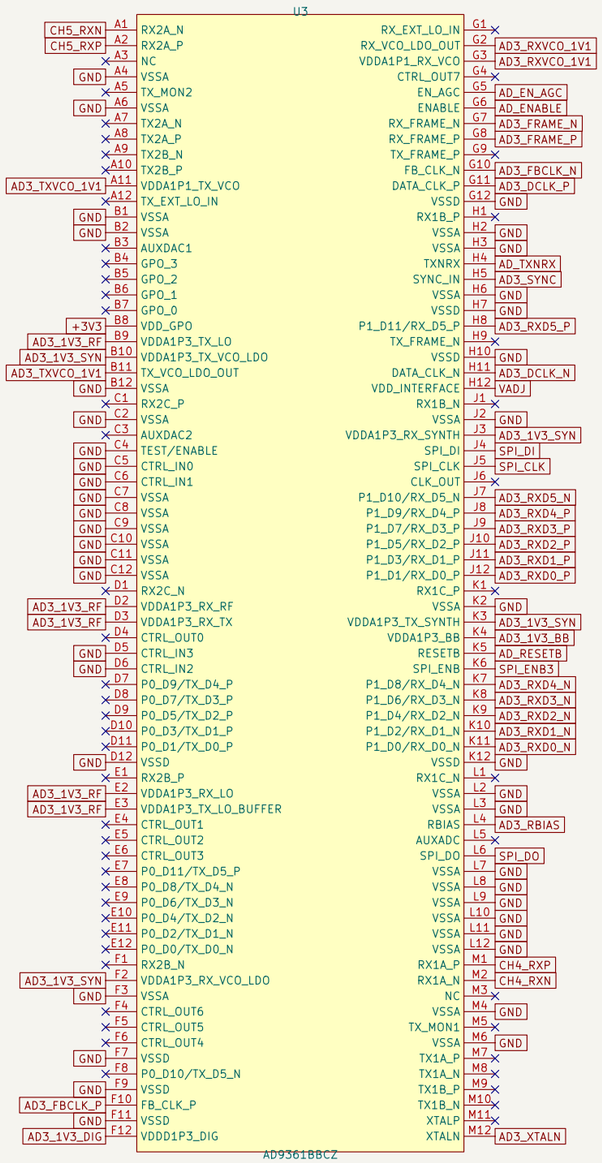
Sheet: /AD9361 #2 (CH2, CH3)/
File: ad9361_2.kicad_sch

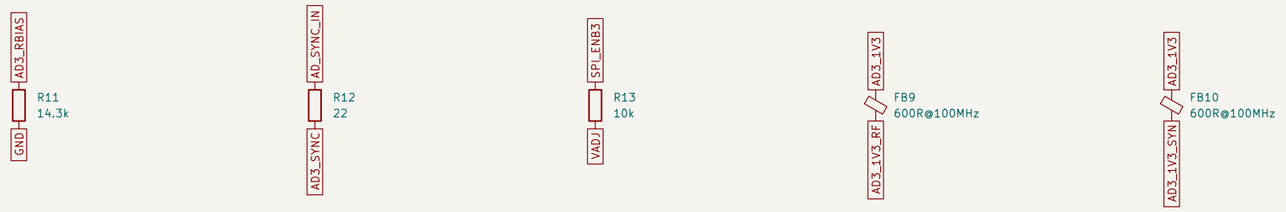
Title: AD9361 #2 (CH2, CH3)

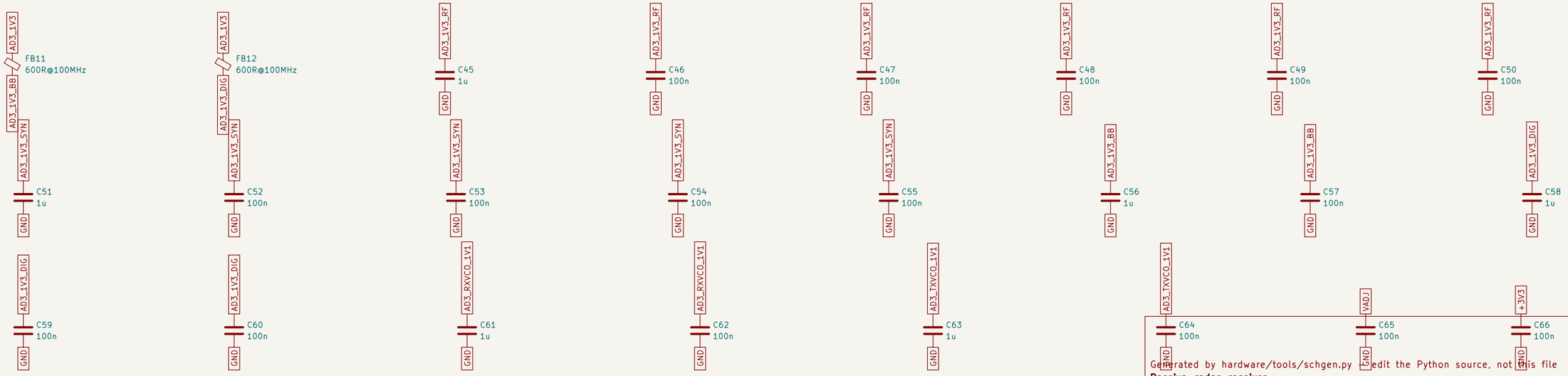
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 3/8

AD9361 #3 (CH4, CH5)

Pin-strap/unused-pin handling follows ADI UG-570 and the AD-FMCOMMS2 reference design; verify.







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Passive radar receiver

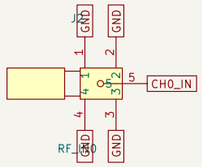
Sheet: /AD9361 #3 (CH4, CH5)/
File: ad9361_3.kicad_sch

Title: AD9361 #3 (CH4, CH5)

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 4/8

RF inputs: SMA, attenuator positions, baluns

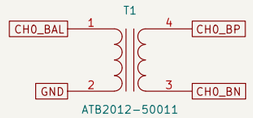
CH0 = reference antenna. Route each P/N pair as 100 ohm differential, equal lengths for all 6 channels.







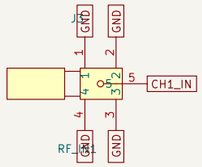








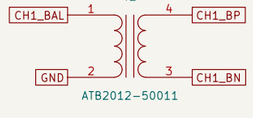








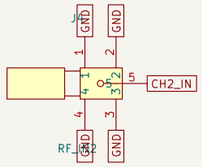








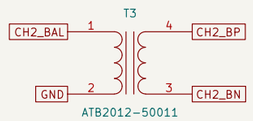








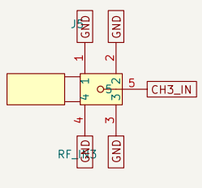








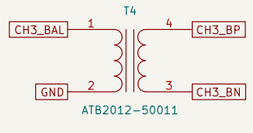








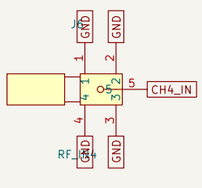








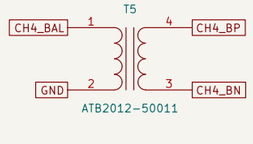








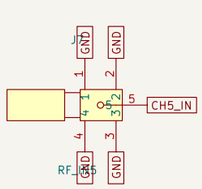




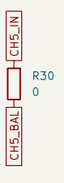




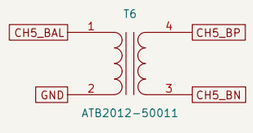


















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Passive radar receiver

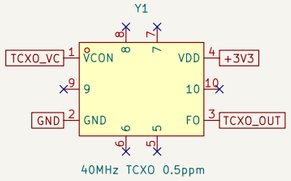
Sheet: /RF inputs: SMA, attenuator positions, baluns/
File: rf_inputs.kicad_sch

Title: RF inputs: SMA, attenuator positions, baluns

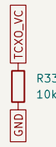
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 5/8

40 MHz reference distribution

All three XTALN traces must be length-matched (< 1 mm). FPGA_REFCLK -> FMC CLK0_M2C_P.



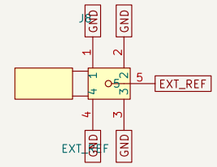






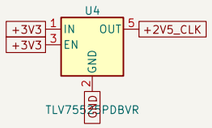








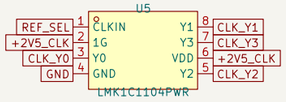










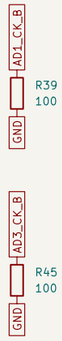






















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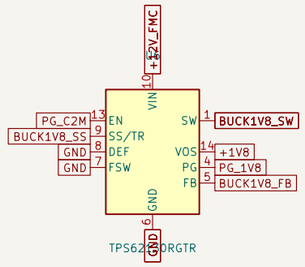
Passive radar receiver

Sheet: /40 MHz reference distribution/
File: clock.kicad_sch

Title: 40 MHz reference distribution

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 6/8

Power: 12 V -> 1.8 V -> 3 x 1.3 V low-noise LDOs













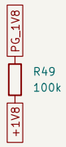


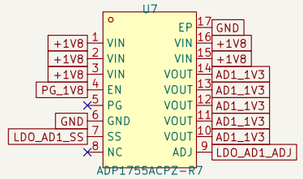












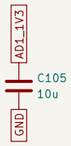






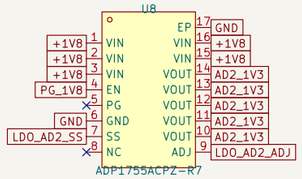


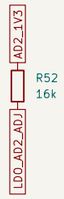


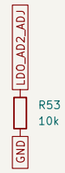














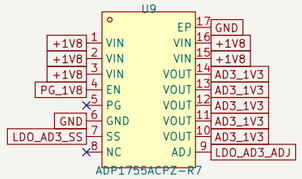


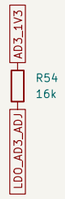
























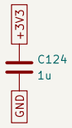






























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Passive radar receiver

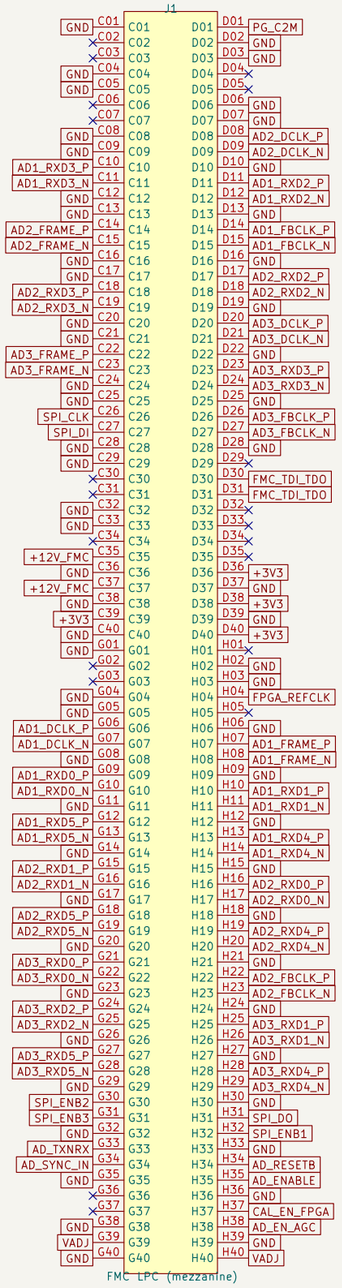
Sheet: /Power: 12 V -> 1.8 V -> 3 x 1.3 V low-noise LDOs/
File: power.kicad_sch

Title: Power: 12 V -> 1.8 V -> 3 x 1.3 V low-noise LDOs

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 7/8

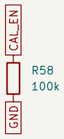
FMC LPC mezzanine connector

VADJ must be set to 2.5 V by the carrier (LVDS_25 banks). LA pair plan: tools/fmc.py.









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Passive radar receiver

Sheet: /FMC LPC mezzanine connector/
File: fmc.kicad_sch

Title: FMC LPC mezzanine connector

Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.3		Id: 8/8